

6) construct the PDA for $L = \{ w c w^R \mid w \text{ is in } (a+b)^* \}$

$L = \{ \epsilon, a b a c a b a, a b c b a, \dots \}$

a | b | c | b | a | ϵ

b |
a |
z₀ |

$$\delta(q_0, a, z_0) = (q_0, a z_0)$$

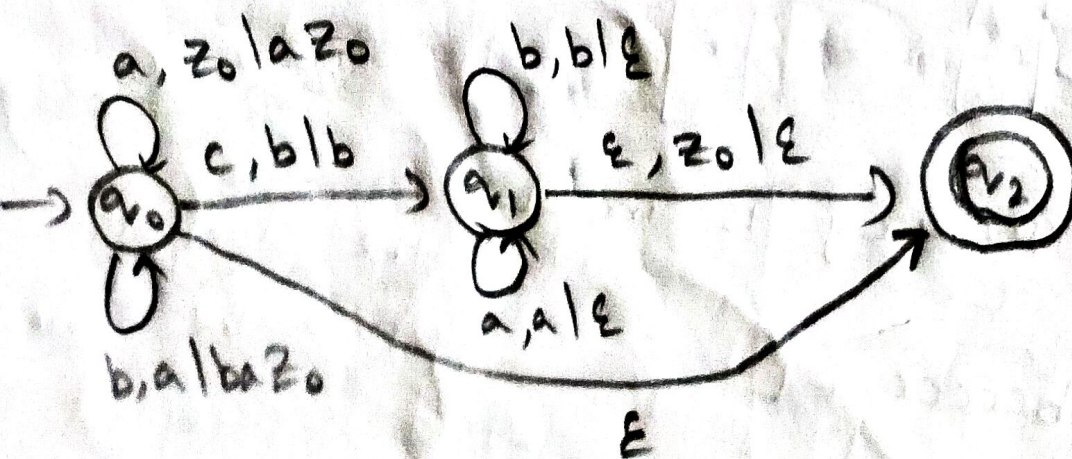
$$\delta(q_0, b, a) = (q_0, b a z_0)$$

$$\delta(q_0, c, b) = (q_1, b)$$

$$\delta(q_1, b, b) = (q_1, \epsilon)$$

$$\delta(q_1, a, a) = (q_1, \epsilon)$$

$$\delta(q_1, \epsilon, z_0) = (q_2, \epsilon)$$



7) Deduce PDA for the given CFL, and test whether 010^4 is acceptable by this PDA.

$$S \rightarrow OBB$$

$$B \rightarrow 0s | 1s | 0$$

The PDA is

$$M = (Q, \Sigma, \Gamma, \delta, q_0, S, F)$$

$Q = \{q\}$ - set of states

$\Sigma = \{0, 1\}$ - input alphabet

$\Gamma = \{S, B, 0, 1\}$ - stack alphabet

$q \rightarrow$ initial state

$S \rightarrow$ initial stack symbol

$F = \emptyset$ - no final states are needed because we accept by empty stack

Transitions for the CFL productions (Non-Terminals):-

Production

$$S \rightarrow OBB$$

$$B \rightarrow 0s | 1s | 0$$

PDA transition

$$\delta(q, \epsilon, S) = (q, OBB)$$

$$\delta(q, \epsilon, B) = \{(q, 0s), (q, 1s), (q, 1)\}$$

Transitions for matching terminals :-

$$\delta(q, 0, 0) = (q, \epsilon)$$

$$\delta(q, 1, 1) = (q, \epsilon)$$

Tracing the string $[w = 010000]$

$$S \Rightarrow OBB \quad [S \rightarrow OBB]$$

$$\Rightarrow 01sB \quad [B \rightarrow 1s]$$

$$\Rightarrow 010BBB \quad [S \rightarrow OBB]$$

$$\Rightarrow 010000 \quad [B \rightarrow 0, B \rightarrow 0, B \rightarrow 0]$$

checking the string acceptance:-

input

stack

•

[S]

→

$\begin{array}{c} a \\ B \\ B \end{array}$

[S → aBB]

x

$\begin{array}{c} x \\ B \end{array}$

→

$\begin{array}{c} x \\ S \\ B \end{array}$

[B → xS]

•

$\begin{array}{c} a \\ B \\ B \\ B \end{array}$

[S → aBB]

•

$\begin{array}{c} a \\ B \\ B \end{array}$

[B → a]

•

$\begin{array}{c} a \\ B \end{array}$

[B → a]

•

$\begin{array}{c} a \end{array}$

[B → a]

8) Give a CFG for the language $N(M)$ where

$M = (\{q_0, q_1\}, \{0, 1\}, \{z_0, x\}, \delta, q_0, z_0, \emptyset)$

where δ is given by,

$$\delta(q_0, 1, z_0) = \{(q_0, xz_0)\}$$

$$\delta(q_0, 1, x) = \{(q_0, xx)\}$$

$$\delta(q_0, 0, x) = \{(q_1, x)\}$$

$$\delta(q_0, \epsilon, z_0) = \{(q_0, \epsilon)\}$$

$$\delta(q_1, 1, x) = \{(q_1, \epsilon)\}$$

$$\delta(q_1, 0, z_0) = \{(q_0, z_0)\}$$

i) Find variable (Non-terminals)

$$CFG, G = (V, T, P, S)$$

$$|V| = Q^2 \Gamma + 1$$

$$= 2^2 (2) + 1$$

$$= 8 + 1$$

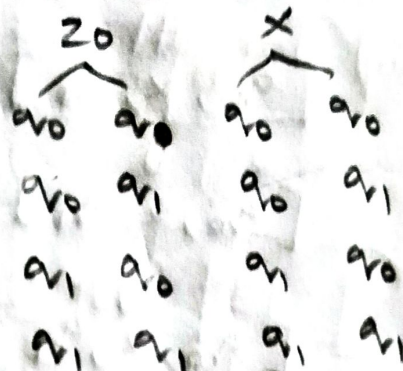
$$= 9$$

$$Q \rightarrow \{q_0, q_1\}$$

$$\Gamma \rightarrow \{z_0, x\}$$

$\Gamma \rightarrow$ stack symbol

$$M = \{ S, \overset{A}{[q_0, z_0, q_0]}, \overset{B}{[q_0, x, q_0]}, \overset{C}{[q_0, z_0, q_1]}, \overset{D}{[q_0, x, q_1]}, \overset{E}{[q_1, z_0, q_0]}, \overset{F}{[q_1, x, q_0]}, \overset{G}{[q_1, z_0, q_1]}, \overset{H}{[q_1, x, q_1]} \}$$



$$T = \{0, 1\}$$

$$S = \{S\}$$

$$4) S \rightarrow [q_0, z_0, q_0] \mid [q_0, z_0, q_1]$$

i) PUSH

$$i) \delta(q_0, 1, z_0) = (q_0, xz_0)$$

$$\overset{A}{[q_0, z_0, q_0]} \xrightarrow{1} \overset{B}{[q_0, x, q_0]} \overset{A}{[q_0, z_0, q_0]}$$

$$\overset{A}{[q_0, z_0, q_0]} \xrightarrow{1} \overset{D}{[q_0, x, q_1]} \overset{E}{[q_1, z_0, q_0]}$$

$$\overset{C}{[q_0, z_0, q_1]} \xrightarrow{1} \overset{B}{[q_0, x, q_0]} \overset{C}{[q_0, z_0, q_1]}$$

$$\overset{C}{[q_0, z_0, q_1]} \xrightarrow{1} \overset{D}{[q_0, x, q_1]} \overset{G}{[q_1, z_0, q_1]}$$

$$\boxed{\begin{array}{l} A \rightarrow IBA \mid IDE \\ C \rightarrow IBC \mid IDG \end{array}}$$

$$ii) \delta(q_0, I, x) = (q_0, xx)$$

$$[q_0, \overset{B}{x}, q_0] \rightarrow I [q_0, \overset{B}{x}, q_0] [\overset{B}{q_0}, x, q_0]$$

$$[q_0, \overset{B}{x}, q_0] \rightarrow I [q_0, \overset{D}{x}, q_1] [\overset{F}{q_1}, x, q_0]$$

$$[q_0, \overset{D}{x}, q_1] \rightarrow I [q_0, \overset{B}{x}, q_0] [\overset{D}{q_0}, x, q_1]$$

$$[q_0, \overset{D}{x}, q_1] \rightarrow I [q_0, \overset{D}{x}, q_1] [\overset{H}{q_1}, x, q_1]$$

$$\boxed{\begin{array}{l} B \rightarrow IBB \mid IDF \\ D \rightarrow IBD \mid IDH \end{array}}$$

2) POP

$$i) \delta(q_0, \epsilon, z_0) = (q_0, \epsilon)$$

$$(q_0, z_0, q_0) \rightarrow \epsilon$$

$$\boxed{A \rightarrow \epsilon}$$

$$ii) \delta(q_1, I, x) = (q_1, \epsilon)$$

$$(q_1, x, q_1) \rightarrow I$$

$$\boxed{H \rightarrow I}$$

3) Read

$$i) \delta(q_0, 0, x) = (q_1, x)$$

$$[q_0, x, q_0] \rightarrow 0 [q_1, x, q_0]$$

$$[q_0, x, q_1] \rightarrow 0 [q_1, x, q_1]$$

$$\boxed{\begin{array}{l} B \rightarrow 0F \\ D \rightarrow 0H \end{array}}$$

$$ii) \delta(q_1, 0, z_0) = (q_0, z_0)$$

$$[q_1, z_0, q_0] \rightarrow 0 [q_0, z_0, q_0]$$

$$[q_1, z_0, q_1] \rightarrow 0 [q_0, z_0, q_1]$$

$$\boxed{\begin{array}{l} E \rightarrow 0A \\ G \rightarrow 0C \end{array}}$$

9) i) find out whether $L = \{x^n y^n z^n \mid n \geq 1\}$ is a context free or not.

$$L = \{xyz, xxyyzz, xxxxyyzzz, \dots\}$$

$$z = xxxxyyzzz, n=3$$

$$|z| \geq n$$

$$9 \geq 3$$

$$\begin{array}{ccccccc} x & x & x & y & y & y & z & z & z \\ \hline u & v & w & x & & & y & & \end{array}$$

$$i) |vwx| \leq n$$

$$|xyy| \leq 3$$

$$3 \leq 3 \checkmark$$

$$\text{ii) } |vxy| \geq 1$$

$$|xy| \geq 1$$

$$2 \geq 1 \checkmark$$

iii) for all $i \geq 0$, uv^iwx^iy

$$i=0$$

$$xxyyzzz \notin L$$

$\therefore$ The given language $L = \{x^n y^n z^n \mid n \geq 1\}$ is not CFL

ii) Let L be the language $\{ww \mid w \in \{0,1\}^*\}$. Show that this language is not a CFL.

$$L = \{0101, 00, 11, 101101, \dots\}$$

$$w = 0101$$

$$w = 0101$$

$$L = \{0^i 1^j 0^i \mid i, j \geq 1\}$$

$$= 0^i 1^j 0^i \mid i, j \geq 1$$

$$Z = 0101, n=1$$

$$|Z| \geq n$$

$$4 \geq 1 \checkmark$$

$$Z = 0101$$

$$uvwx y$$

$$1) |vwx| \leq 1$$

$$|101| \leq 1$$

$$3 \leq 1 \quad \times$$

$\therefore$ The given Language is not a CFL

What is a closure Property?

A closure property tells us whether a particular type of language remains in the same class after applying an operation to it.

Simple meaning:

If we take a Context-Free Language and perform an operation on it, and the result is also a CFL, then we say:

CFLs are closed under that operation.

Closure Under Union :-

If L_1 and L_2 are CFLs then:

$L_1 \cup L_2$ is also a CFL

Proof:-

~~$$L_1 = \{a^n b^n \mid n \geq 0\}$$~~

~~$$L_2 = \{a^n c^n \mid n \geq 0\}$$~~

~~Both are CFLs~~

~~$$L_1 \cup L_2$$~~

$$L_1 = \{a^n \mid n \geq 0\}$$

$$L_2 = \{b^n \mid n \geq 0\}$$

$$L_1 = \{\epsilon, a, aa, aaa, \dots\}$$

$$L_2 = \{\epsilon, b, bb, bbb, \dots\}$$

$$L_1 \Rightarrow S_1 \rightarrow aS_1 \mid \epsilon$$

$$L_2 \Rightarrow S_2 \rightarrow bS_2 \mid \epsilon$$

$$L_1 \cup L_2 \Rightarrow S \rightarrow aS_1 \mid bS_2 \mid \epsilon$$

$$\Rightarrow S_1$$

$$\Rightarrow aS_1$$

$$\Rightarrow aaS_1$$

$$\Rightarrow aa$$

Concatenation :- [Concatenation means joining a string from L_1 with a string from L_2

$$L_1 = \{a^n \mid n \geq 0\}$$

$$L_2 = \{b^n \mid n \geq 0\}$$

$L_1 \cdot L_2$ is also CFL

$$L_1 \cdot L_2 \rightarrow S_1 \cdot S_2$$

$$S \rightarrow S_1 \cdot S_2$$

$$S \rightarrow aS_1, bS_2 \mid \epsilon$$

$$L_1 = \{a, aa\}$$

$$L_2 = \{b, bb\}$$

$$L_1 \cdot L_2 = \{ab, aabb, abb, aab\}$$

Closure under Kleene star :-

If L is a CFL then

L^* is also a CFL

Kleene star means zero or more repetitions of strings in the language

Ex:

$$L = \{ab\}$$

Then

$$L^* = \{\epsilon, ab, abab, ababab, \dots\}$$

$$S \rightarrow ab$$

$$S \rightarrow S, S \mid \epsilon$$

abS

ababS

$\therefore L^*$ is a CFL

Closure under Reversal :-

Statement :-

If L is a CFL then

L^R is also a CFL

Reversal means writing every string in the opposite direction.

for example:-

$w = abc$

then

$w^R = cba$

If

$L = \{ab, abc\}$

then

$L^R = \{ba, cba\}$

$A \rightarrow abc$

$A \rightarrow cba$

Closure Under Intersection with a Regular Language

If L is a CFL and R is a regular language

then

$L \cap R$

is CFL

A CFL can be recognized by a Pushdown Automaton

A regular language can be recognized by a finite Automaton (FA).

We can combine

PDA for L

FA for R

to create a machine that accepts only strings accepted by both machines.

Therefore: $L \cap R \rightarrow$ is CFL

$CFL \cap Regular = CFL$

It does not mean:

$CFL \cap CFL = CFL$ is false in general.

~~$\{a,b\}$~~ $\{0,1\}$

$S_1 \rightarrow \bullet S_1$, $S_1 \rightarrow 1S_1$

$\{0,00,000\}$ $\{1,01,011\}$